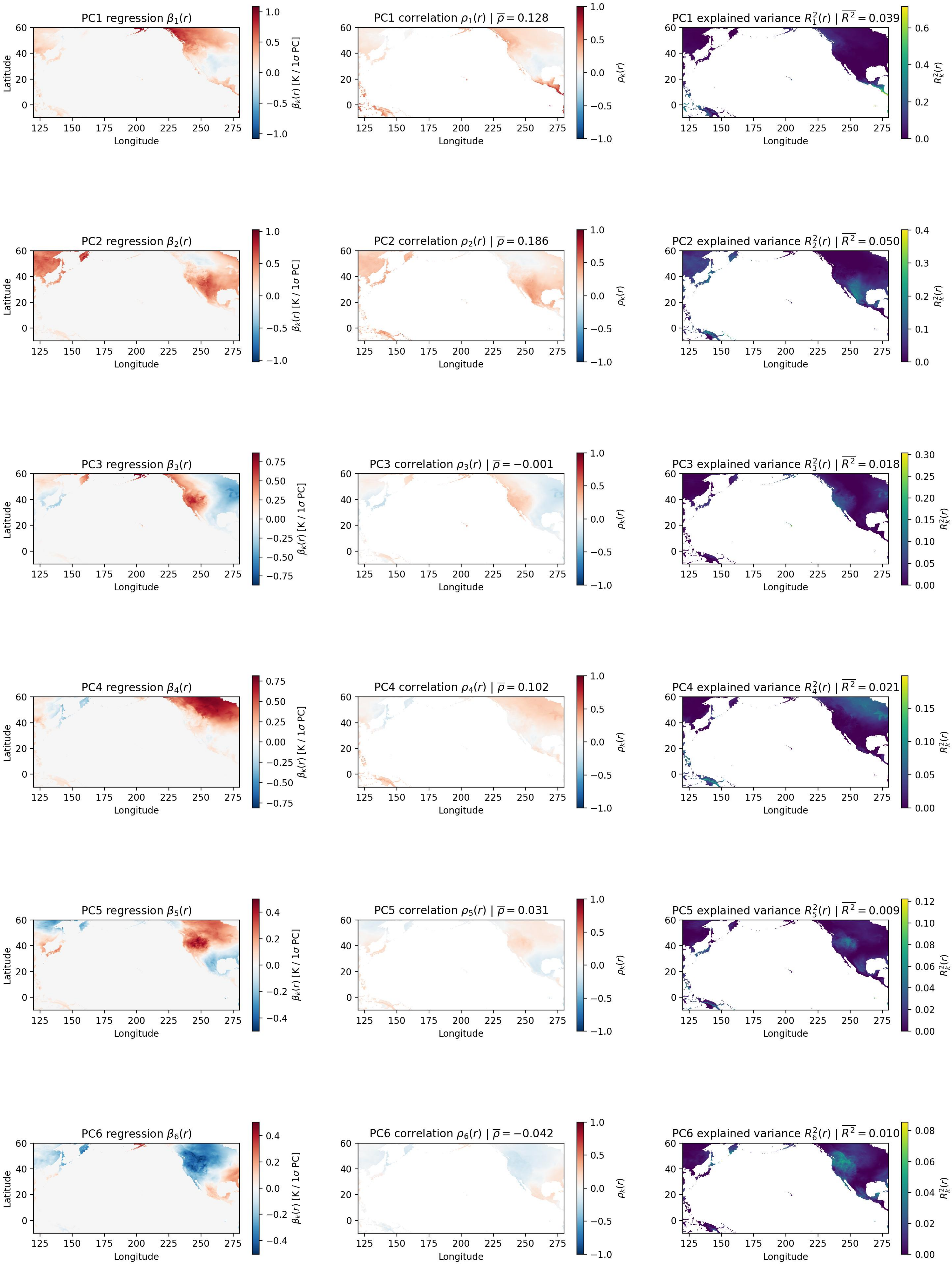

$\beta_k(r)$ is the physical T2m response pattern over the matched Pacific ERA5-Land region.

$\beta_k(r) = \frac{\sum_t a_k(t)Y(t,r)}{\sum_t a_k(t)^2}$, where $Y(t,r)$ is the ERA5-Land T2m anomaly at time t and matched-region land grid point r, and $a_k(t)$ is COBE2 Pacific SST PC k at time t.

PCs are standardized here, so $\beta_k(r)$ is in K per one-standard-deviation increase in PCk.

k: PC index, for example $k = 1$ means PC1. t: time index, for example one month or one season. r: one matched-region ERA5-Land grid point.

$\rho_k(r) = \mathrm{corr}(a_k(t), Y(t,r))$ is the correlation between COBE2 Pacific SST PC k and matched-region T2m at grid point r.

$R_k^2(r) = \rho_k(r)^2$ is the fraction of local matched-region T2m variance explained by perfect knowledge of SST PC k.